

True soft proofing relies on the gamut of the display or desktop printer being larger than the gamut of the printing press.

This allows the colour engine in the colour management software to shift and constrain the image colour values inside the confines of the display or printer gamut so that the image appears as it will when printed on the press.

Out Of Gamut Colours

As mentioned in a previous feature because the gamut of most printers is smaller than that of capture devices such as cameras and scanners, some colours may be lost when converting from the captured image into the print device colour space (see also Intents). Colour editing software usually has a setting which will highlight out of gamut colours.



These three images of an audio CD cover design demonstrate soft proofing with the out of gamut warning turned on. Out of gamut colours are shown by the green highlight. The original image is on the left, the proof for printing that image on a desktop printer is in the middle and the proof for a commercial press is on the right

Comparing Soft Proofs With Prints

In practice, when final printed output is compared side by side with the screen, soft proofing to a display often appears to be only partially successful. This is because, assuming the display has been calibrated and profiled correctly, the image will be displayed with the chosen white point (for example 6,500K) at a certain brightness (often 80 cd/m2) while also being affected to some extent by the ambient illumination reflected from the screen. In contrast the hard copy proof will be illuminated by the available light and this will almost certainly not be of the same colour temperature as the screen, nor will the proof reflect at the same brightness.

For the best correspondence between display soft proofs and final output the display should be set up alongside a viewing booth, with both the display and the booth illuminated by the same fairly low level of neutral ambient light. The colour temperature, and brightness of the display must be adjusted to be as close as possible. If the booth has a brightness control this may be the easiest way to balance the brightness to that of the display.

The display colour temperature should be set to match that of the lamps in the booth;

5,000K for D50 lamps and 6,500K for D65. Unfortunately the stated CT of viewing booth lamps isn't particularly accurate. It may be necessary to set the monitor CT to something other than one of the standard settings to match the booth exactly and to re-calibrate and re-profile the display for this setting.

Desktop Printers And Colour Management

Desktop colour printers, particularly inkjet are designed as 'RGB' printers. The operating system sends print data as RGB and the printer driver and printer perform the necessary conversions to print the image correctly using CMYK inks. When images are printed from an image editing application that provides user controls for colour management a choice must be made over whether either the colour management in the application, or in the printer driver should be used. It is a common mistake to use both, which results in unpredictable colours in the final print. Only very recently have printer drivers started to include warnings indicating that both management systems may be in use.

The best approach is to select the correct printer profile in the colour controls for the image editor and to turn off the colour management in the printer driver. Unfortunately in many desktop printer drivers it is often difficult to determine exactly how to disable their internal colour and image controls. For example in the current Epson driver it is necessary to select Printer Preferences, click the Advanced button, select ICM and tick the Off (No color adjustment) box.



Selecting ICM and ticking the Off box, on the Advanced page of the Epson print preferences driver control panel, turns off colour management by the driver

Conclusion – Defining Your Work Flows

Perhaps the best way to approach colour management is to first define the work flows you will use.

For example a photographer might only ever work with bitmap images and will mainly work in RGB up until the printing stage. If these are to be only small run prints rather than printed as part of a publication on a commercial press then is no need to prepare separations, most of the work flow can be in RGB and the CMYK conversions either left to the desktop printer and its driver. A hobbyist photographer might choose to set their camera for sRGB output. Depending on the camera sRGB might be the only choice. A professional photographer is more likely to choose RAW mode, a mode which outputs the image data as registered by the CCD sensor, without any intermediate processing by the camera electronics. Using this mode often requires the use of a dedicated RAW mode driver supplied by the camera manufacturer.

A graphics designer will often import digital photographs, but will manipulate and combine these with vector graphic elements. Graphic design work is usually intended for publication via a commercial printing press so it is sensible to choose to work with the effects in CMYK (this is CorelDRAWs default mode) and to soft proof using the commercial press profile.

Article References

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